

FOUR USEFUL FINITE MIXTURE MODELS FOR REGRESSION ANALYSES OF PANEL DATA WITH A CATEGORICAL DEPENDENT VARIABLE

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This paper describes and contrasts two useful ways to employ a latent class variable as a mixture variable in regression analyses of panel data with a categorical dependent variable. One way is to model unobserved heterogeneity in the trajectory, or change in the distribution, of the dependent variable. Two models that accomplish this are the latent trajectory model and latent growth curve model for a categorical dependent variable having ordered categories. Each latent class here represents a distinct trajectory of the dependent variable. The latent trajectory model introduces covariate effects on the composition of latent classes, while the latent growth curve model introduces covariate effects on both the “intercept” and the “slope” of growth in logit, which may vary among latent classes.

The other useful way is to model unobserved heterogeneity in the state dependence of the dependent variable. Two models that accomplish this are introduced for a simultaneous analysis of response probability and response stability, and the latent class variable is employed to distinguish two latent populations that differ in the stability of responses over time. One of them is the

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switching multinomial logit model with a time-lagged dependent variable as its separation indicator, and the other is the mover-stayer regression model.

By applying these four models to empirical data, this paper demonstrates the usefulness of these models for panel-data analyses. Example programs for specifying these models based on the LEM program are also provided.

1. INTRODUCTION

I describe here two useful ways to employ a latent class variable as a mixture variable in regression analyses of panel data with a categorical dependent variable, and it introduces four models, two for each of the two ways. The description includes only finite mixture models and does not include other models that employ latent class variables for the analysis of panel data. I also restrict the description of models to those with a categorical dependent variable to narrow the scope of coverage. Although these models are useful in revealing distinct latent patterns of change in the categorical dependent variable, three of them—all except the latent growth curve model, which I include for comparative purposes—have not been introduced in the literature, at least in the regression form presented in this article. They are thus novel in that sense, although the derivation of these models owes much to some of the preceding studies reviewed below.

A general model that employs latent class variables for the analysis of panel data with a categorical dependent variable is the mixed Markov latent class model (van del Pol and Langeheine 1990), which has more recently called the mixture latent Markov (MLM) model (Vermunt, Tran, and Magidson 2006). By assuming both discrete states and discrete time, this model further assumes (1) *latent states* among which transitions occur over time, while the states are reflected by some observable indicators, and (2) *latent classes of persons*, each having a distinct transition probability pattern. The first latency reflects a correction for *measurement errors* for discrete states among which transitions occur; it assumes that the true states are not observable and are reflected only by their indicators. The second latency reflects *unobserved population heterogeneity* in the transition probability matrix; it assumes that there are latent groups of persons with different patterns of transitions among discrete states (Yamaguchi 2002; Vermunt et al. 2007).

Generally, parameters of an MLM model cannot be estimated without setting some strong constraints. As a result, two special cases of the MLM model, each of which omits one aspect of latency, are more often considered. One is the mixture Markov (MM) model, described by Poulsen (1982) and Langeheine and van de Pol (1990). This model assumes that the discrete states among which transitions occur are observed and not latent, while assuming latent classes of persons each of which has a distinct transition probability matrix. The MM model thus ignores measurement errors but incorporates unobserved population heterogeneity in transition probabilities. The other special case of the MLM model is the latent Markov (LM) model introduced by Wiggins (1955), Langeheine (1988), and van de Pol and De Leeuw (1989). Models employed by Collins (1991; Collins and Wugalter 1992) in the "latent transition analysis" and elaborated for the analysis of drug-use progression are also LM models. This model assumes latent states with their indicators observed at each time and a single transition probability matrix among those latent states. The LM model thus ignores unobserved population heterogeneity in the transition probabilities, while incorporating a correction for measurement errors for states among which transitions occur.

A model is referred to as a finite "mixed" model rather than a finite "mixture" model if a latent class variable for unobserved population heterogeneity in a two-level model of time and person is introduced only as an independent source of variability in the dependent variable among people, as in the case of the Heckman-Singer model for hazard rates (Heckman and Singer 1974a, 1974b). On the other hand, a model is referred to as a finite mixture model if a latent class variable is employed for modeling unobserved population heterogeneity in the structure of the relationship among variables, including the dependent variable. The MM model, which assumes unobserved heterogeneity in the transition probabilities of the dependent variable, is thus a finite mixture model, while the LM model is neither a mixed nor a mixture model. The latent class membership is time constant in the MM model, while it is time varying in the LM model.

This paper introduces four special cases of the mixture Markov model by formulating them as regression models. Compared with the relatively widely used latent Markov models, the general mixture Markov model has been used much less often in empirical research mainly because, in its general form, its parameter estimates are not yet stable. The

four specific models introduced in this paper, however, can attain stable parameter estimates by making each of them reflect unobserved population heterogeneity in one of the two structural aspects in the change pattern of $Y(t)$ over time, as described below.

As is well known in log-linear and related models for social mobility, the structure of cross-classification of $Y(t)$ and $Y(t - 1)$ for a categorical variable Y includes two components of change: (1) change in the marginal distribution of $Y(t)$ over time, which is called the “structural mobility” in mobility analysis, and (2) the dependence of $Y(t)$ on $Y(t - 1)$, characterized by log odds ratios independently of change in the marginal distribution of $Y(t)$, which is called “circulation mobility” in mobility analysis (Sobel, Hout, and Duncan 1985). For simplicity, these two elements will be referred to in this paper as the “trajectory” and the “state dependence” components of change in $Y(t)$. A general mixture Markov model tries to characterize unobserved heterogeneity for both of these components simultaneously, and I consider this a major cause of the lack of stable parameter estimates for the model because data often do not have information rich enough for identifying both aspects of unobserved heterogeneity simultaneously. Hence, two distinct types of more specific mixture Markov models can theoretically be considered: (1) models that characterize unobserved heterogeneity only in the “state dependence” in $Y(t)$ and (2) models that characterize unobserved population heterogeneity only in the “trajectory” of $Y(t)$.

Two distinct models are introduced in the first category, both of which are novel in the literature at least in a regression form with covariates. A latent class variable in these models characterizes distinct latent patterns in the dependence of $Y(t)$ on $Y(t - 1)$. While the models permit a time-constant effect of the latent class variable on $Y(t)$, the baseline pattern of change in $Y(t)$ as a function of time, or trajectory, is assumed to be independent of latent classes. As a latent class model, this model satisfies *local dependence* because indicators $\{Y(t)\}$ are not independent, controlling for latent classes.

The mixture Markov models for modeling unobserved heterogeneity in state dependence are useful for a simultaneous analysis of response probability and response stability, because the models can employ a latent class variable to characterize heterogeneity in the stability of responses over time. This paper introduces two such models, and one of them, which will be referred to as the *mover-random switching regression model* (the “mover-random model” for simplicity), is a

switching regression model for a categorical dependent variable with the time-lagged dependent variable $Y(t - 1)$ as its separation indicator. A separation indicator of switching regression is an instrumental variable assumed to affect the outcome in one latent class but not in the other. Since the model assumes that $Y(t)$ depends on $Y(t - 1)$ in one latent class but not in the other, and since we can regard the extent of dependence of $Y(t)$ on $Y(t - 1)$ as an indicator of response stability, the distinction of the two latent classes represents unobserved heterogeneity in the stability of responses over time. The model consists of a pair of regression equations, one for predicting $Y(t)$ and the other for predicting latent class membership. This model can be considered a regression extension of the mover-random mixture model introduced by Clogg, Rudas, and Xi (1995) for the analysis of mobility tables.

We shall refer to another mixture Markov model for modeling unobserved heterogeneity in state dependence as the *mover-stayer regression model*. For a dichotomous dependent variable $Y(t)$, such as a variable with “positive” and “negative” responses, this model assumes two latent classes of “stayers,” one for positive response and the other for negative response, each of which is assumed to be the same over time with probability 1, as well as a third latent class, “movers,” whose responses are subject to a Markov process. The model consists of a pair of regression equations, one for predicting $Y(t)$ among movers and the other for predicting the probability of being a stayer.

A special case of this mover-stayer model, the stayer-random model, is a regression extension of the “black-white (BW) model” (Converse 1964), whose formulation as a log-linear quasi-independence model (Goodman 1968, 1975) was introduced by Taylor (1983) and Duncan, Stenback, and Brody (1988) for the analysis of three-wave panel data without covariates. The mover-stayer model introduced below not only extends the BW model into a regression model with covariates but also relaxes its assumption by permitting first-order Markov state dependence for the latent class of movers. Hazard rate models that employ the mover-stayer assumption also exist (Yamaguchi 1992; Kuk and Chen 1992), and they are known in medical statistics as “cure-rate models” (Li and Taylor 2002).

Two distinct models will also be introduced in the second category—that is, models in which a latent class variable characterizes unobserved population heterogeneity in the trajectory of $Y(t)$. We refer to them as the latent trajectory model and the latent growth curve

model. While the two models are identical in their basic structure, they differ, as described below, in the use of covariates. Assuming a dependent variable with ordered categories, the models incorporate an assumption that yields a hierarchical order in the average level of $Y(t)$ among latent classes at each given time. These models typically assume local independence among a set of $Y(t)$ s, which are the indicators of the latent variable. In other words, they typically assume no state dependence, or independence of $Y(t)$ from $Y(t - 1)$, for each latent class.¹ This assumption is made to yield stable parameter estimates because latent classes are usually identified by assuming that association among $Y(t)$ s results from unobserved heterogeneity.

Latent trajectory analysis was introduced by Nagin and Land (1993; Land, McCall, and Nagin 1996; Land, Nagin, and McCall 2001) on the basis of a semiparametric mixture of Poisson processes for the analysis of event count data, but a related model is formulated in this article for the analysis of a categorical dependent variable. The latent growth curve model is typically applied to a continuous dependent variable (Muthen 2001) but can also be applied to a categorical dependent variable with ordered states (Vermunt and van Dijk 2001). The latent trajectory and latent growth curve models for a categorical dependent variable described in this paper are not fundamentally new; the first one is a special case of the multinomial logit latent-class regression model (Bandein-Roche et al. 1997; Yamaguchi 2000), and the second one is a variant of the nonparametric latent growth curve model introduced by Vermunt and van Dijk (2001). A major difference between the two models is that in the latent trajectory model, covariate effects are introduced to characterize differences in the composition of latent classes, as in the mixture model of Poisson processes introduced by Nagin and Land (1993) for the analysis of event count data. On the other hand, in the

¹ We may introduce, as an extension of these models, a parameter, or a set of parameters, which characterize the dependence of $Y(t)$ on $Y(t - 1)$ as long as the extended model assumes that such state dependence is independent of latent classes, so that the latent class variable does not confound the modeling of unobserved population heterogeneity in trajectory with that in association. The simplest example is a model that assumes time-constant uniform association between two adjacent time points of $Y(t)$. However, the nature of latent classes changes because they are identified without using information on the extent of time-constant association between $Y(t)$ and $Y(t - 1)$. We do not consider those extensions in this paper, however.

latent growth curve model, covariate effects are introduced to characterize differences in growth curves and possible interaction effects between covariates and the latent class variable on the “intercept” and “slope” of the growth curve. The modeling of missing data for both latent trajectory and latent growth curve models will also be described, because it is very typical to have a missing-data problem in latent trajectory and latent growth curve analysis.

Two illustrative applications will also be presented, an application of the first two models for response stability and an application of the last two models for latent trajectory and latent growth curve analysis. Sample LEM programs for these applications are also given.

2. MOVER-RANDOM SWITCHING REGRESSION MODEL

This model assumes a categorical dependent variable Y and a dichotomous latent class variable. One of the two latent classes represents a Markov process in which $Y(t)$ depends on $Y(t-1)$, and the other represents a Bernoulli process in which $Y(t)$ is independent of $Y(t-1)$, in order to let the two latent classes represent a difference in the stability of responses over time measured by the strength of association between $Y(t)$ and $Y(t-1)$. While it is possible to relax this assumption to make two latent classes represent two Markov processes with different extents of association between $Y(t)$ and $Y(t-1)$, parameter estimates of such a model are usually unstable without an alternative instrumental variable as an effective separation indicator of latent classes.² The model below employs the time-lagged Y , $Y(t-1)$, as the separation indicator. As shown in the example, this works very well for making parameter estimates stable. For the same reason, an extension into models with three or more latent classes will require some additional instrumental variables to attain stable parameter estimates.

This model assumes two regression equations: (1) a multinomial logit equation for the analysis of response probabilities of Y and

² However, if no state-dependence effects exist and the dependent variable has ordered categorical states, the switching multinomial logit model where a distinct intercept, time effects, and covariate effects may exist for each latent class may converge well, if data are sufficiently rich, without an instrumental variable. This paper introduces such a switching regression model for the analysis of a categorical dependent variable with ordered states below as the latent growth curve model.

(2) a logistic equation for the composition of two latent classes. Let $\pi_i^{Y(t)|Y(t-1)} \equiv \text{Prob}(Y(t) = i | Y(t-1))$ be the conditional probability of $Y(t) = i$ for a given value of $Y(t-1)$, let D_{ii}^Y , for $i = 1, \dots, l-1$, be the set of $l-1$ variables for the deviation contrast of $Y(t)$, let X be a dichotomous latent-class variable that takes values of 1 and 0, and let $\pi^X \equiv \text{Prob}(X = 1)$. The model can then be specified by the following pair of equations:

$$\log(\pi_i^{Y(t)|Y(t-1)} / \pi_1^{Y(t)|Y(t-1)}) = a_{ii} + b_i X + \sum_{j=1}^{I-1} c_{ij} D_j^{Y(t-1)} X + \mathbf{d}'\mathbf{V}(t) \quad (1)$$

$$\log[\pi^{X|Y(1)} / (1 - \pi^{X|Y(1)})] = \alpha + \sum_{j=1}^{I-1} \gamma_j D_j^{Y(1)} + \beta' \mathbf{U} \quad \text{for } i = 2 \sim I, t = 2 \sim T. \quad (2)$$

In equation (1), covariates $\mathbf{V}(t)$ may or may not be time dependent and may include time-constant covariates $\mathbf{U}$ included in equation (2) as a subset. Component $\sum_{j=1}^{I-1} c_{ij} D_j^{Y(t-1)} X$ in equation (1) represents the interaction effects of X and $Y(t-1)$ on $Y(t)$, when the effects of $Y(t-1)$ on $Y(t)$ are characterized by the set of category-by-category interactions involving $(I-1)^2$ parameters. The component, $\sum_{j=1}^{I-1} c_{ij} D_j^{Y(t-1)}$, however, may be replaced, especially when Y is a variable having ordered categories, by a simpler form of interaction such as uniform association (Goodman 1979). An important characteristic of equation (1) is that while the interaction effects of X and $Y(t-1)$ on $Y(t)$ exist, there are no main effects of $Y(t-1)$ on $Y(t)$. This is because we impose the conditions that when $X = 1$, $Y(t)$ depends on $Y(t-1)$, but when $X = 0$, $Y(t)$ is independent of $Y(t-1)$.

In equation (2), covariates $\mathbf{U}$ are time constant, and the component $\sum_{j=1}^{I-1} \gamma_j D_j^{Y(1)}$ indicates that the latent class composition may depend on the time-1 response of Y to reflect the *content dependence* of response stability, such that the extent of response stability, measured by the probability of being in latent class 1, may differ depending on whether, for example, the response was “positive,” “neutral,” or “negative” at time 1.

In equations (1) and (2), the deviation contrast is employed for $Y(t)$. While this specification is not essential in equation (2), it is important in equation (1) because the effect of X (coefficient b) then reflects the average effect of X on $Y(t)$. It is also normally the case that the

model's parameter estimates converge more rapidly, and the iterative search for the maximum likelihood estimates is more likely to be successful if the deviation contrast for $Y(t)$ is used than if the indicator contrast (through a set of dummy variables) is used in this model.

The model of equations (1) and (2) can be extended in a straightforward manner in several ways. The first-order Markov assumption of equation (1) for one latent class may also be relaxed to permit further dependence on the past. As is typical in the switching regression model, we can test whether covariate effects differ between the two latent classes. Such a hypothesis can be tested by introducing an interaction effect of X and $V(t)$ in equation (1). Similarly, we can test whether the content dependence of response stability also varies with covariates by introducing the interaction effects of $Y(1)$ and U in equation (2).³

On the other hand, we should not include the interaction effects of $Y(t-1)$ and a time-constant variable V in equation (1) when V is a covariate in equation (2). This is because information in data on the interaction effect of $Y(t-1)$ and V on $Y(t)$, which is an alternative way to characterize the dependence of the extent of association between $Y(t)$ and $Y(t-1)$ on V , overlaps greatly with information on the effect of V on X ,⁴ and the primary aim of this model is to identify the covariate effects on response stability, which is characterized by the distinction of the two latent classes. The interaction effect of a time-dependent covariate $V(t)$ and $Y(t-1)$ in equation (1) is, however, useful in assessing the variation of stability with a time-dependent covariate.

The pair of equations is greatly simplified if the dependent variable Y is also a dichotomous variable with values 1 and -1 (by the deviation contrast) such that

³ However, if $Y(1)$ and $Y(t)$ for $t > 1$ are strongly associated in the data, a simultaneous inclusion of both an interaction effect of X and U on $Y(t)$ in equation (1) and an interaction effect of $Y(1)$ and U on X in equation (2) may make these two effects highly correlated and unstable.

⁴ One may question why we do not simply model the response stability by introducing the interaction effects of $Y(t-1)$ and V on $Y(t)$. There are a few substantive reasons in addition to a statistical reason that alternative models always help to determine which model fits the data better. The first is that a presentation of results by a pair of regression results enables the effects on stability to be described separately from the effects on response probability. Second, the content dependence of response stability can be assessed in a more straightforward manner in the proposed models, especially when the content dependence also interacts with covariates. Third, the proposed models provide useful descriptive statistics about the variation among covariate states in the proportion of people having stable attitudes.

$$\log [\pi_t^{Y(t)|Y(t-1)} / (1 - \pi_t^{Y|Y(t-1)})] = a_t + bX + cXY(t-1) + \mathbf{d}'\mathbf{V}(t); \quad (1a)$$

$$\log [\pi^{X|Y(1)} / (1 - \pi^{X|Y(1)})] = \alpha + \gamma Y(1) + \beta'\mathbf{U}. \quad (2a)$$

3. MOVER-STAYER REGRESSION MODEL

This section considers a model that assumes a dichotomous dependent variable for simplicity—although the model can be extended for a categorical variable having three or more categories by adding more categories for stayers. For a dichotomous dependent variable, the model assumes that there are three latent classes, and two of them are for the classes of “stayers,” one for the positive response and the other for the negative response, for which the response made at time 1 is repeated at later times with probability 1. The third latent class consists of “movers,” who can change their responses over time but may repeat the same response by chance. If we further assume that $Y(t)$ does not depend on $Y(t-1)$ among movers, we have a regression extension of the “black-white model” formulated as a log-linear quasi-independence model by Taylor (1983) and Duncan et al. (1988) without covariates—where “being black” implies people with a strong conviction about their responses such that they do not change their responses over time and “being white” implies people without any conviction about their responses such that their responses are independent over time. If we regard the Markov state dependence as a “gray” state in between the two, the mover-stayer model described below can be regarded as a “black-gray model,” while the mover-random model described in the previous section is a “gray-white model.” Technically, we may also have a “black-gray-white” model, and parameters of such a model can be identified for a dependent variable with three or more categories for which we can distinguish between the “diagonal” state dependence among stayers and state dependence resulting from an “off-diagonal” association of $Y(t)$ and $Y(t-1)$ among “movers.” However, the “black-gray-white” model is very likely to be unstable in estimating the proportion of the “gray” population for dichotomous dependent variables, because the “gray” population can then be approximated well by a mixture of the “black” and “white” populations.

A regression extension of the mover-stayer model assumes a pair of probabilities $[\pi^{Y(t)|Y(t-1)}, \pi^{X|Y(1)}]$ at the individual level. Here, $\pi^{X|Y(1)}$ is the conditional probability that a subject, for a given value of $Y(1)$, will be a stayer (for whom $X = 1$ holds), thereby repeating at later times the same response as that at time 1, rather than a mover (for whom $X = 0$ holds). $\pi^{Y(t)|Y(t-1)}$ is the conditional probability of making a positive rather than a negative response, at time t , for $t \geq 2$, for a mover when his response at time $t - 1$ is given. Hence, $\pi^{Y(t)|Y(t-1)}$ is defined only for movers for whom $X = 0$ holds and should be denoted more precisely as $\pi^{Y(t)|Y(t-1), X=0}$. Then, for the three time points of observations, for example, the joint conditional probability $P_{ab|c}$ for $Y_2 = a$ and $Y_3 = b$ for a given $Y_1 = c$, where $a, b, c = 1$ or 0 , for a subject takes the following form

$$\begin{aligned} P_{11|1} &= \pi_{X1|1} + (1 - \pi_{X1|1})\pi_{Y(2)|1}\pi_{Y(3)|1}; \\ P_{10|1} &= (1 - \pi_{X1|1})\pi_{Y(2)|1}(1 - \pi_{Y(3)|1}); \\ P_{01|1} &= (1 - \pi_{X1|1})(1 - \pi_{Y(2)|1})\pi_{Y(3)|0}; \\ P_{00|1} &= (1 - \pi_{X1|1})(1 - \pi_{Y(2)|1})(1 - \pi_{Y(3)|0}); \\ P_{11|0} &= (1 - \pi_{X0|0})\pi_{Y(2)|0}\pi_{Y(3)|1}; \\ P_{10|0} &= (1 - \pi_{X0|0})\pi_{Y(2)|0}(1 - \pi_{Y(3)|1}); \\ P_{01|0} &= (1 - \pi_{X0|0})(1 - \pi_{Y(2)|0})\pi_{Y(3)|0}; \\ P_{00|0} &= \pi_{X0|0} + (1 - \pi_{X0|0})(1 - \pi_{Y(2)|0})(1 - \pi_{Y(3)|0}). \end{aligned}$$

This defines the joint likelihood of (Y_2, Y_3) .

Suppose that we also code the dependent variable $Y(t)$ by a dummy variable having the values of 1 and 0. The pair of regression equations, one for logit of $\pi^{Y(t)|Y(t-1)}$ and the other for logit of $\pi^{X|Y(1)}$, can then be specified as follows

$$\log[\pi^{Y(t)|Y(t-1)} / (1 - \pi^{Y(t)|Y(t-1)})] = a_t + b Y(t-1) + \mathbf{c}'\mathbf{V}(t) \quad \text{for } X = 0; \quad (3)$$

$$\log[\pi^{X|Y(1)} / (1 - \pi^{X|Y(1)})] = \alpha + \gamma Y(1) + \beta'\mathbf{U}. \quad (4)$$

In equation (3), covariates $\mathbf{V}(t)$ may or may not be time dependent and may include time-constant covariates $\mathbf{U}$ included in equation (4) as

a subset. Component $bY(t - 1)$ in equation (3) represents the Markov state dependence among moves, and component $\gamma Y(1)$ in equation (4) represents the content dependence of response probability. When we set $b = 0$ in equation (3), we obtain the regression extension for the black-white model, or the stayer-random model. Not unlike the case of the mover-random model described in the previous section, this model should not include the interaction effect of covariate V with $Y(t - 1)$ on $Y(t)$ in equation (3) when V is a covariate in equation (4), because the data may not be able to separate effectively the large state-dependence effect from the probability of being a stayer. The major advantage of using the mover-stayer model is its ability to identify covariate effects on response stability measured by the probability of being a stayer. The interaction effect of a time-dependent covariate $V(t)$ and $Y(t - 1)$ in equation (3) is, however, useful in assessing the variation of stability with a time-dependent covariate.

4. ADJACENT LOGIT LATENT-TRAJECTORY MODEL

We assume that the dependent variable $Y(t)$ has the ordered states below. As stated in the introduction, the model described below, which will be referred to as the *adjacent logit latent trajectory model*, is a special case of the multinomial logit latent class regression model (Bandein-Roche et al. 1997; Yamaguchi 2002), where the latent class variable becomes the dependent variable for regression with covariates. However, there are three specific elements that are unique to this latent class regression model. First, $Y(t)$, for $t = 1, \dots, T$, are the indicators of the latent class variable, and each latent class represents a distinct trajectory of $Y(t)$. This is the same as the latent-trajectory model for event count data (Nagin and Land 1993; Land et al. 1996, 2001) as a mixture of Poisson processes. Second, since the membership of latent classes is assumed to be time invariant, only time-constant covariates are allowed for the latent class composition, and a time-dependent variable is assumed to affect $Y(t)$ directly. This is also the same as the Nagin-Land model. Third, while the trajectory is characterized in the Nagin-Land model for event count data by time-varying effects of the latent class variable on the mean of $Y(t)$ plus time effects on the mean of $Y(t)$, this characteristic is modified in the adjacent logit model, because, unlike the analysis of event count data, the dependent variable in the model is a

categorical variable that does not have a mean. The modification is that in order to make the effects of the latent class X on $Y(t)$ characterize systematic differences among latent classes in the level of Y , we impose the proportional odds assumption regarding the effect of X on $Y(t)$ for each discrete time t when the categories of $Y(t)$ are properly ordered so that the effect of the latent class variable on the adjacent logit of $Y(t)$ is uniform, while we allow these proportional effects of X on $Y(t)$ to change nonlinearly with time.

For the main time effects that apply to all latent classes, however, the proportional odds assumption is not made. Such an assumption would cause the marginal distributions of $Y(t)$ not to be fixed. However, the latent class model with categorical indicators is usually very unstable without fixing the marginal distribution of the indicators. Hence, the model introduces unconstrained threshold-specific nonlinear main time effects on Y to force the expected marginal distribution of $Y(t)$ at each time t be identical to its observed distribution.

More formally, the model is characterized by the pair of equations shown below. Let $\pi_{ij}^{Y_i} \equiv \text{Prob}(Y(t) = j)$, let X be the latent class variable, let D_m^X be a dummy variable that takes a value of 1 if $X = m$, and let $\pi_m^X \equiv \text{Pr ob}(X = m)$. Then

$$\log(\pi_{ij}^{Y_i} / \pi_{i(j+1)}^{Y_i}) = a_{ij} + \sum_{m=2}^M b_{im} D_m^X + \mathbf{c}' \mathbf{V}(t); \quad (5)$$

$$\log(\pi_m^X / \pi_1^X) = \alpha_m + \beta'_m \mathbf{U}. \quad (6)$$

In equation (5), parameters a_{ij} fix the marginal distributions of $Y(t)$, and we impose the proportional odds assumption that the effects $\{b_{im}\}$ of X on the log odds of adjacent categories j and $j + 1$ of $Y(t)$ do not depend on threshold j , although the effects change freely with time to reflect a distinct trajectory for each latent class. This proportional odds assumption permits us to regard a particular latent class L_1 to be at a higher, the same, or a lower level of Y than another class L_2 at each time t , depending on whether $b_{tL_1} - b_{tL_2}$ is positive, zero, or negative. Note that the standard assumption of local independence among indicators—that is, $\pi^{XY_1 Y_2 Y_3 Y_4} = \pi^X \pi^{Y_1|X} \pi^{Y_2|X} \pi^{Y_3|X} \pi^{Y_4|X}$, where $\pi^{XY_1 Y_2 Y_3 Y_4}$ is the joint probability of the five variables—is implicit in equation (5) from the fact that there is no direct association among $Y(t)$ s within each latent class.

Unlike the models described in the preceding sections, the model of equations (5) and (6) does not include common covariates. We employ time-constant covariates in equation (6) because the latent class membership is constant over time, and we include only time-dependent covariates in equation (5). However, an alternative formulation is introduced in the next section.

Panel data frequently involve the problem of missing data. Generally, the indicators $Y(t)$ of the latent class variable may not be observed consistently at different times for everybody in the sample. In particular, the missing mechanism of $Y(t)$ may depend on the latent-trajectory pattern itself such that some latent classes may be more likely than others to drop out of the sample, thereby generating nonignorable nonresponse (Little and Rubin 2002). The distinction between missing completely at random (MCAR) and missing at random (MAR) does not affect the estimates of parameters in equations (5) and (6); it affects only the estimates of parameters that characterize the marginal distribution of the missing indicator described below and its dependence on observed covariates. Hence, if we can assume MAR, we can impose MCAR without loss of generality for the latent trajectory analysis based on this model.

However, the situation is different if the missing data are not missing at random. In log-linear analysis, nonignorable nonresponse can be modeled as a latent class model (Winship and Mare 1989), and it is easy to extend it to a model that includes another latent class variable for the purpose of latent trajectory analysis. In particular, we can test a hypothesis that the missing mechanism of $Y(t)$ depends on latent trajectory patterns.

For example, suppose that we have four time points of observations for $Y(t)$, but some observations have missing data for $Y(4)$. Then we have two data sets of cross-classifications $Y(1)Y(2)Y(3)Y(4)$, and $Y(1)Y(2)Y(3)$. Let R , where $R = 1$ or 2 , be the indicator of this missing-data pattern. Then the model is MCAR if R is independent of all $Y(t)$ s and is MAR if R depends only on $Y(1)$, $Y(2)$, and $Y(3)$, but not $Y(4)$. On the other hand, we have nonignorable nonresponse (NIR) if R depends on $Y(4)$ or if R depends on the latent class variable X with $Y(1)$ through $Y(4)$ as its indicators.

Let $Y(4)^* = Y(4)$ if $R = 1$, and $Y(4)^* = L$, where L is a latent variable for unobserved response of $Y(4)$, if $R = 2$. Then we obtain a modified equation by replacing $Y(4)$ with $Y^*(4)$ in equation (5). A model that hypothesizes NIR such that the occurrence of missing data

depends on the latent trajectory of $Y(t)$ can be specified by adding a third equation:

$$\log(\pi^R/(1 - \pi^R)) = d + \sum_{m=2}^M e_m D_m^X, \quad (7)$$

where $\pi^R = \text{Prob}(R = 1)$. Models that specify various MAR hypotheses can also be tested by using an alternative third equation that specifies the dependence of $\text{logit}(\pi^R)$ on observed $Y(t)$ s. These models can be generalized into models with more than two variables with missing data. The NIR model that specifies equation (7) as well as models that specify various MAR conditions can be easily tested by LEM, as described later for an application.

5. ADJACENT LOGIT LATENT GROWTH CURVE MODEL

The model described below for $Y(t)$ having ordered states is a variant of the latent growth curve model introduced by Vermunt and van Dijk (2001) and is described mainly for the comparative purpose of formulating the model in the same modeling framework as that of the latent trajectory model. Unlike the model described by Vermunt and van Dijk, however, the model below (1) assumes unconstrained threshold-specific nonlinear main time effects to attain the stability of parameter estimates and (2) extends the model to a switching regression model where covariate effects on the “intercept” and “slope” of growth may vary among latent classes.

For the basic model without covariates, we assume a model that is identical to the adjacent logit latent trajectory model described in the preceding section. Although the intercept is separated from the time effect in typical growth curve models, we retain parameters a_{ij} in equation (8) below because we wish to fix the marginal distribution of the indicators of the latent class variable in order to obtain stable parameter estimates, as stated in the preceding section. When covariates are introduced, we first include not only time-dependent covariates but also time-constant covariates in the equation for predicting $Y(t)$ rather than X such that

$$\log(\pi_{ij}^{Y_i}/\pi_{i(j+1)}^{Y_i}) = a_{ij} + \sum_{m=2}^M b_{im} D_m^X + \mathbf{c}'\mathbf{U} + \mathbf{d}'\mathbf{V}(t); \quad (8)$$

$$\log(\pi_m^X/\pi_1^X) = \alpha_m, \quad (9)$$

where $\mathbf{U}$ and $\mathbf{V}(t)$ in equation (8) are time-constant and time-dependent covariates, respectively.

In equation (8), however, the shape of the trajectory does not depend on covariates. In particular, we may hypothesize that the *conditional* “growth rate,” or the “slope” of linear growth in logit, *controlling for the main nonlinear time effects* a_{ij} , may depend on $\mathbf{U}$ by hypothesizing the interaction effect of linear time and $\mathbf{U}$ such that

$$\log(\pi_{ij}^{Y_i}/\pi_{i(j+1)}^{Y_i}) = a_{ij} + \sum_{m=2}^M b_{im} D_m^X + \mathbf{c}'\mathbf{U} + \mathbf{d}'\mathbf{V}(t) + \mathbf{e}'\mathbf{U}t. \quad (8a)$$

These models are multilevel models, because in addition to the built-in multinomial error term at the time level, they have an additional error term with a discrete distribution at the individual level. A further elaboration of the model of equation (8a) to make the covariate effects on the slope of growth to be nonlinear is certainly possible.

Suppose that we also introduce the interaction effects of $\mathbf{U}$ and X in equation (8a) such that

$$\begin{aligned} \log(\pi_{ij}^{Y_i}/\pi_{i(j+1)}^{Y_i}) = & a_{ij} + \sum_{m=2}^M b_{im} D_m^X + \mathbf{c}'\mathbf{U} + \mathbf{d}'\mathbf{V}(t) + \mathbf{e}'\mathbf{U}t \\ & + \sum_{m=2}^M \mathbf{f}'_m \mathbf{U} D_m^X. \end{aligned} \quad (8b)$$

The effect of D_m^X is then given as $b_{im} + \mathbf{f}'_m \mathbf{U}$, and, therefore, not only does the intercept of trajectory vary with covariates $\mathbf{U}$, but this variation also differs among latent classes. This model is a switching regression model, because each latent class may have a distinct set of regression coefficients for covariate effects.

We may further elaborate this switching regression model by hypothesizing that the covariate effects on the “slope” of the growth curve also vary with latent classes such that

$$\begin{aligned} \log(\pi_{ij}^{Y_i} / \pi_{i(j+1)}^{Y_i}) = & a_{ij} + \sum_{m=2}^M b_{im} D_m^X + \mathbf{c}'\mathbf{U} + \mathbf{d}'\mathbf{V}(t) + t\mathbf{e}'\mathbf{U} \\ & + \sum_{m=2}^M \mathbf{f}_m' \mathbf{U} D_m^X + \sum_{m=2}^M t\mathbf{g}_m' \mathbf{U} D_m^X. \end{aligned} \quad (8c)$$

The effect of D_m^X is then given as $b_{im} + (\mathbf{f}_m' + t \mathbf{g}_m') \mathbf{U}$, and, therefore, not only do the intercept and the slope of growth both vary with covariates $\mathbf{U}$, but their variations also differ among latent classes.

What is the information in data that identifies a variation with covariates in the “intercept” and the “slope” of the growth curve among latent classes? Note first that when we hypothesize an interaction effect of X and U on $Y(t)$, we hypothesize that the relationship between the latent class variable and its indicators varies with U . Since the characteristics of latent classes are determined by their association with their indicators, the dependence of this association on U implies that we actually have many more distinct latent trajectories than the number of categories M of the latent class variable and that differences among the sets of M latent trajectory classes vary with U . The strength of association between X and $Y(t)$ under the assumption of local independence of $Y(t)$ measures the strength of association among the $Y(t)$ s in the data, which is a measure of response stability, indirectly. Hence, the interaction effects of U and X on $Y(t)$ reflect a variation of response stability with U such that, other things being equal, subjects in a state of U that shows a stronger association of X and $Y(t)$ have greater response stability. Hence, although the latent growth curve model does not aim at analyzing response stability directly, the model utilizes information on the dependence of response stability on covariates indirectly in identifying the interaction effects of X and covariates on $Y(t)$.

The model selection requires more elaboration in the latent growth curve model than in the latent trajectory model. For the application of the latent trajectory model, the analysis can be done in two steps: (1) identify the number of latent classes in the model without covariates and (2) introduce covariates in predicting the composition of latent classes, because the introduction of covariates for the composition of latent classes hardly affects the optimal number of latent classes. The situation, however, is different in the latent growth curve model. Since covariate effects on both the “intercept” and the “slope” of growth introduce more observed heterogeneity in the pattern of trajectory, less

unobserved population heterogeneity in the trajectory of $Y(t)$ may remain. Hence, the optimal number of latent classes must be identified for each given model of covariate effects.

6. METHOD OF PARAMETER ESTIMATION

The maximum likelihood method is used for parameter estimation for all the models described in this paper. The LEM program (Vermunt 1997a, 1997b) is employed for the application of all the models. Sample LEM specifications of models employed in the following analysis are given in Appendix A.

7. APPLICATION 1: TWO MODELS FOR THE ANALYSIS OF RESPONSE STABILITY

7.1. Data and Variables

For an illustrative application of the two models for the analysis of response stability described in earlier sections, we employ the data from the American National Election Study (ANES), 1990–1992. The population consists of men and women aged 20–69 years old. This three-wave panel survey covers the period from the Persian Gulf War to the time before Bill Clinton's victory over George H. W. Bush in the 1992 presidential election. The ANES data show a dramatic change in the approval rate for Bush's presidency, from 65.7 percent in 1990, to 79.5 percent in 1991 after the recovery of Kuwait, to 41.8 percent in 1992 before the election.

The dependent variable is a dichotomous variable describing whether the subject approved of Bush's presidency measured in 1991 and 1992 (Y_2 and Y_3). Explanatory variables are all categorical variables and, in addition to Y_1 , include (1) age group in 1991 (five categories), (3) race (two categories), (3) education (four categories), (4) religious preference (three categories), and (5) party preference in 1990 (three categories). Descriptive statistics of these covariates are given in Table 1.

7.2. Analyses

Table 2 presents main results regarding models' goodness-of-fit with data. The results show that for both the mover-random model and

TABLE 1
Distributions of Covariate States: $N = 839$

	Proportion (%)
1. Age group	
20–29	22.5
30–39	27.8
40–49	21.1
50–59	12.7
60–69	16.0
2. Race	
Whites	90.1
Blacks	9.9
3. Education	
Less than 12 years	12.7
12 years	37.1
13–15 years	24.3
16 years or more	25.9
4. Religious preference	
Protestant	44.2
Catholic	26.0
Other religion	15.4
None	14.4
5. Party preference	
Republican	26.7
Democrat	38.6
Other, none	34.7

the mover-stayer model, models II-A and II-B which hypothesize heterogeneity in response stability (by introducing covariate effects on $\text{logit}(\pi^X)$) are significantly better than models I-A and I-B, respectively, which hypothesize homogeneous response stability (by assuming the intercept only for $\text{logit}(\pi^X)$), and that models III-A and III-B, which hypothesize further that interaction effects of party preference and time-1 response $Y(1)$ on $\text{logit}(\pi^X)$ exist, are significantly better than models II-A and II-B, respectively. In addition, the results show that among models that hypothesize heterogeneous response stability, the stayer-random models that hypothesize independence of responses over time among movers (models II-B2 and III-B2) are better, or more parsimonious in the use of parameters, than the more general mover-stayer models II-B1 and III-B1, respectively.

Table 3 presents results of parameter estimates from mover-random model III-A, mover-stayer model III-B1, and stayer-random

TABLE 2
Main Results

	np	2*LL	p
I. Models without heterogeneity in stability			
A. Mover-random model	18	-10,946.5	
B. Mover-stayer models			
B1. Mover-stayer	17	-10,950.5	
B2. Stayer-random	16	-10,959.3	
II. Models with heterogeneity in stability (main effects only)			
A. Mover-random model	32	-10,904.7	
B. Mover-stayer models			
B1. Mover-stayer	31	-10,913.1	
B2. Stayer-random	30	-10,916.1	
III. Models that add the interaction effect of party preference and Y(1) on $\text{logit}(\pi^X)$			
A. Mover-random model	34	-10,890.3	
B. Mover-stayer models			
B1. Mover-stayer	33	-10,900.6	
B2. Stayer-random	32	-10,901.3	
Model II-A vs. I-A	14	41.8	<0.001
Model II-B1 vs. I-B1	14	37.4	<0.001
Model II-B2 vs. I-B2	14	43.2	<0.001
Model II-B1 vs. II-B2	1	3.0	>0.05
Model III-A vs. II-A	2	14.4	<0.001
Model III-B1 vs. II-B1	2	12.5	<0.01
Model III-B2 vs. II-B2	2	14.8	<0.001
Model III-B1 vs. III-B2	1	0.7	>0.5

model III-B2.⁵ See Tables A1 and A2 in Appendix A for the programming of the first two models by the use of LEM. For each model, the first column presents the covariate effects for the logistic regression on the response probability of $Y(t)$ and the second column presents the

⁵ The log-likelihood (LL) value given in the output of LEM for the data of cross-classified frequencies is the value of the model for the contingency table even if the model applied specifies a dependent variable with multinomial responses. To obtain the LL value for a regression model with multinomial responses, we subtract $\sum_c f_c \log(f_c/N)$ from the LL value of the LEM output, where c represents each combination of covariates' states and f_c is the observed frequency of combination c —which indicates in the present case a combined state of $Y(1)$, age group, education, race, religious preference, and party preference. N is the total frequency. The LL values reported in Table 2 are those of the LEM output. The value of $\sum_c f_c \log(f_c/N)$ in the present case is $-4,731.60$.

TABLE 3
Results from Two Models for the Analysis of Response Stability

	Mover-Random Model (Y coded 1 vs. -1)		Mover-Stayer Model (Y coded 1 vs. 0)		Stayer-Random Model (Y coded 1 vs. 0)	
	logit ($\pi^{Y(t)}$)	logit (π^X)	logit ($\pi^{Y(t)}$)	logit (π^X)	logit ($\pi^{Y(t)}$)	logit (π^X)
1. Age group in 1991 (vs. 20-29)						
30-39	0.035	1.018*	0.067	0.896*	0.093	0.842*
40-49	0.018	0.594	0.005	0.592	0.024	0.543
50-59	0.110	0.493	0.201	0.428	0.265	0.397
60-69	-0.166	1.421**	-0.362	1.259**	-0.364	1.213***
2. Race (vs. nonblacks)						
Blacks	-0.645***	0.671	-1.280***	0.278	-1.305***	0.349
3. Education (vs. 12 years)						
<12 years	-0.176	-0.130	-0.177	-0.331	-0.165	-0.286
13-15 years	-0.190	0.487	-0.375	0.406	-0.381	0.421
16+ years	-0.569***	1.300***	-0.781**	0.736*	-0.755**	0.759**
4. Religious preference (vs. Protestants)						
Catholics	-0.160	0.681	-0.213	0.412	-0.198	0.407
Other	-0.124	-0.128	-0.196	-0.229	-0.227	-0.193
Religion						
None	-0.570***	0.864*	-1.324***	0.719*	-1.367***	0.759*
5. Party preference (vs. Republican)						
Democrat	-0.608**	-0.140	-1.163**	1.396*	-1.132**	1.398*
Other, none	-1.094***	0.908	-1.667***	1.051	-1.631***	1.148
6a. Dependence on the previous year's response						
Y(t - 1)	—	—	0.408	—	—	—
6b. Dependence on latent class X						
X	-0.546	—	—	—	—	—
6c. Interaction effects of Y(t - 1) and X						
X*Y(t - 1)	2.198***	—	—	—	—	—
7a. Dependence on the first year's response Y(1)						
	—	0.584	—	1.328	—	1.304
7b. Interaction effects of Y(1) and party preference (vs. Republican)						
Democrats	—	-1.306*	—	-2.965**	—	-3.005***
Other, none	—	0.264	—	-0.968	—	-1.124
8a. Y-Intercept						
1991	2.129***	—	3.827***	—	4.173***	—
1992	0.412	—	0.332	—	0.656	—
8 X-Intercept						
	—	-2.276***	—	-2.775***	—	-2.672***

covariate effects on the logit of being in the first latent class, whose response is more stable than that of the second latent class. The difference among the three models is that the contrast of the first and the second latent classes is the comparison of the state-dependent and state-independent responses in the mover-random model, the comparison of being a stayer and being a mover in the mover-stayer model, and the comparison of being a stayer and being a “random-response person” in the stayer-random model. Note also that the response variable $Y(t)$ is based on the use of deviation contrasts of 1 versus -1 in the mover-random model, while the states of $Y(t)$ are coded 1 and 0 in the mover-stayer and stayer-random models. This coding difference thus needs to be taken into account in interpreting regression coefficients in terms of log-odds ratio.

Note that since the stayer-random model omits from the mover-stayer model a parameter for the dependence of $Y(t)$ on $Y(t-1)$ among movers and is a better model than the mover-stayer model two things hold: (1) the covariate effect of $Y(t-1)$ on $\text{logit}(\pi^{Y(t)})$ is not significant in the results of the mover-stayer model, and (2) the effects of all other covariates are qualitatively the same in terms of the presence or absence of significant effects and the direction of significant effects between the two models. However, the deletion of the effect of $Y(t-1)$ on $\text{logit}(\pi^{Y(t)})$ in the stayer-random model tends to yield smaller standard errors for covariate effects on response stability (on $\text{logit}(\pi^X)$). Hence, when the stayer-random model fits the data equally well, it is recommended that this model, rather than a more general mover-stayer model, be used, in order to make a more powerful analysis of response stability. The results from the mover-random and stayer-random models are described and compared below.

The race effects in Table 3 indicate that blacks are less likely to have a positive attitude toward the Bush administration than whites, and there are no race effects on response stability. These findings are consistent among the three models. Numerically, blacks have 0.275 [$=\exp(-0.645 \times 2)$] times as great odds of having a positive as of having a negative attitude in the results of the mover-random model. The corresponding odds ratio is 0.271 [$=\exp(-1.305)$] in the results of the stayer-random model. The effects of educational attainment indicate that, compared with those with 12 years of education, those with 16 or more years of education have not only a more negative attitude toward the Bush administration but also a higher response stability. Similarly, the effects

of religious preference indicate that, compared with Protestants, those who have no religious preference have not only a more negative attitude toward the Bush administration but also a higher response stability. The two models are consistent regarding these two sets of findings.

Regarding the effects of party preference, both models consistently show that, compared with Republicans, Democrats and Other/None are strongly more negative toward the Bush administration. The effects of party preference on response stability depend on whether the response is positive or negative, because of the significant presence of interaction effects of party preference and the time-1 response on the latent class composition. In the results from the mover-random model, where $Y(1)$ is coded 1 versus -1 , Democrats have less stable attitudes than Republicans when the time-1 response is positive, with an odds ratio of 0.236 [$= (\exp(-0.140 - 1.306))$] for the odds of the stable-response class versus the unstable-response class. On the other hand, Democrats have more stable attitudes than Republicans when the time-1 response is negative, with an odds ratio of 3.21 [$= \exp(-0.140 + 1.306)$]. In the results from the stayer-random model, where $Y(1)$ is coded 1 versus 0, Democrats have less stable attitudes than Republicans when the time-1 response is positive, with an odds ratio of 0.201 [$= (\exp(1.398 - 3.005))$] for the odds of the stable-response class versus the unstable-response class. On the other hand, Democrats have more stable attitudes than Republicans when the time-1 response is negative, with an odds ratio of 4.05 [$= \exp(1.398)$]. Hence, for both models, the results show that the “status-consistent” attitudes—namely, the Republicans’ positive attitude and the Democrats’ negative attitude toward the Republican presidency—are more stable than the “status-inconsistent” attitudes—namely, the Republicans’ negative attitude and the Democrats’ positive attitude toward the Republican presidency. Results from the two models are consistent regarding the age effects on response probability and response stability as well.

Hence, the results from the two models are qualitatively very similar. This is because the stayer-random model virtually holds. This fact is not only indicated by an absence of the effect of $Y(t - 1)$ on $Y(t)$ among movers in the results of the mover-stayer model, it is also strongly suggested by an extremely high degree of dependence of $Y(t)$ on $Y(t - 1)$ in latent class 1, with an odds ratio of 6323.32 [$= \exp(2.198 \times 4)$], in the results of the mover-random model. The goodness of fit of the two models is also close, as seen in the similar value of log-likelihood in Table 2.

TABLE 4
Average Probability of Being in Each Latent Class

	Mover-Random		Mover-Stayer		Stayer-Random	
	Mover	Random	Stayer	Mover	Stayer	Random
I. Overall average probability of being in each latent class	0.436	0.564	0.353	0.647	0.368	0.632
II. Average probability of being in each latent class by group						
1. Age group						
20–29	0.312	0.688	0.240	0.760	0.260	0.740
30–39	0.517	0.483	0.413	0.587	0.425	0.575
40–49	0.408	0.592	0.343	0.657	0.359	0.641
50–59	0.384	0.616	0.307	0.693	0.323	0.677
60–69	0.548	0.452	0.456	0.544	0.469	0.531
2. Race						
Whites	0.433	0.567	0.352	0.648	0.365	0.635
Blacks	0.467	0.533	0.361	0.639	0.393	0.617
3. Education						
Less than 12 years	0.352	0.648	0.248	0.752	0.266	0.734
12 years	0.349	0.651	0.300	0.700	0.311	0.689
13–15 years	0.437	0.563	0.374	0.626	0.390	0.610
16 years or more	0.601	0.399	0.460	0.540	0.478	0.522
4. Religious preference						
Protestant	0.396	0.604	0.331	0.669	0.343	0.657
Catholic	0.486	0.514	0.377	0.623	0.388	0.612
Other religion	0.365	0.635	0.274	0.726	0.295	0.605
None	0.543	0.457	0.461	0.539	0.487	0.513
5. Party preference ¹						
Republican	0.413	0.587	0.381	0.619	0.395	0.605
with Y(1) = “P”	0.445	0.555	0.414	0.596	0.429	0.571
with Y(1) = “N”	0.190	0.810	0.140	0.860	0.144	0.856
Democrat	0.336	0.664	0.291	0.709	0.305	0.695
with Y(1) = “P”	0.207	0.793	0.160	0.840	0.175	0.825
with Y(1) = “N”	0.490	0.510	0.450	0.550	0.475	0.525
Other, none	0.566	0.434	0.401	0.599	0.417	0.583

¹“P”: positive response; “N”: negative response.

Table 4 presents the estimates for the average probability of being in each latent class for the three models of Table 3. It is the average because the composition of latent classes depends on the set of covariates that are hypothesized in the model to affect the odds of being in one versus the other latent class.

The results from Table 4 show that the overall proportion, or the average probability, of a more stable class is 0.436 in the mover-random model, 0.353 in the mover-stayer model, and 0.368 in the stayer-random model. Generally, between the mover-random model and the mover-stayer model, we can expect a smaller proportion of a more stable class for the mover-stayer model than for the mover-random model because “stayer” in the mover-stayer model satisfies a more strict condition of stability than “mover” in the mover-random model. Equivalently, we can expect a smaller proportion of a less stable class for the mover-random model than for the mover-stayer model because the “random” population in the mover-random model satisfies a more strict condition of instability than “movers” in the mover-stayer model. The results from Table 4 are consistent with those expectations. We can also expect that the division of population into the “stayers” and “random population” in the stayer-random model will yield middle values in between the results from the mover-random model and the mover-stayer model. This expectation is also confirmed in the results of Table 4 but the table also shows that the results of the stayer-random model are much closer to those of the mover-stayer model than to those of the mover-random model.

More concretely, the results from the stayer-random model, for example, show that groups that are expected to be more stable judging from the covariate effects on logit (π^X) in Table 3, such as those aged 30–39 and 60–69, those with 16 or more years of education, those with no religious preference, and Republican and Democrats with “status consistent” responses at time 1, have higher response stability as indicated by a high estimate for the proportion of stayers (more than 40 percent). An exception is respondents with “Other/None” party preference among whom the estimate for the proportion of stayers is also high. We found in Table 3, however, that the effect of having “Other/None” party preference is positive but not significant. In fact, the attitude of this group is stable owing to compositional effects, because the average probability of being a stayer reflects confounding effects of correlated states. In particular, the proportion of people who have no religious preference is significantly higher (with a 1 percent level of significance) among those with “Other/None” party preference than among Republicans and Democrats, and people with “Other /None” party preference have higher levels of education than Republicans (with a 5 percent level of significance). The results of Table 4 also show that

the attitudes of Republicans and Democrats with “status inconsistent” responses at time 1 are very unstable (with stayers forming less than 20 percent of each group), and people aged 20–29 and those with less than 12 years of education also have relatively smaller response stability (with stayers forming less than 30% of each group).

8. APPLICATION 2: TWO ADJACENT LOGIT LATENT CLASS MODELS FOR PANEL-DATA ANALYSIS

8.1. Data and Variables

As an illustrative application of the two models for latent trajectory and latent growth curve analyses, the trajectory of wives’ marital satisfaction is analyzed. The data are taken from the Panel Survey on Consumer’s Life, 1993–2001, collected by the Research Institute on Household Economics, Japan. The marital satisfaction variable was collected every two years from 1995 and consists of five categories (1 “very satisfied”; 2 “pretty satisfied”; 3 “neither satisfied nor dissatisfied”; 4 “somewhat dissatisfied”; and 5 “not satisfied at all”). The results from the 1995, 1997, 1999, and 2001 waves are included here. The population is the sample of married women aged 24–34 in 1993 and a supplementary sample of married women aged 24–27 in 1997. The sample is also restricted to those whose marital satisfaction was measured at least at three consecutive times among the four durations—namely, the 1st–2nd year, the 3rd–4th year, the 5th–6th year, and the 7th–8th year of marriage—and this yields a sample size of 380. Note that the requirement that three time periods are observed eliminates women who got divorced within 4 years and will generate a sample selection bias.⁶ Hence, the findings about the trajectory of marital satisfaction analyzed below should be qualified by the statement that they apply to women whose marital duration is more than 4 years.

We have the following three response types for missing data in this sample: (1) the ABCD type, which consists of subjects from whom

⁶ I could have reduced this sample selection bias by including subjects for whom there were observations for at least two, rather than three, consecutive time points, and by including AB, BC, and CD types as three additional response types of missing observations.

we have data of $Y(t)$ at all four time points, (2) the ABC type, which consists of subjects from whom we have data of $Y(t)$ only for $t = 1, 2$, and 3, and (3) the BCD type, which consists of subjects from whom we have data of $Y(t)$ only for $t = 2, 3$, and 4. The sample proportions of these three types are 22.9 percent, 31.6 percent, and 45.5 percent, respectively.

8.2. Analyses

Table 5 presents the results from 15 models without covariates, which are a set of combinations of a distinct number of latent classes and a distinct type of missing data mechanism (MCAR, MAR-1, MAR-2, MAR-3, NIR). MAR-1 represents the dependence of the distinction of the ABCD and ABC types on $Y(1)$ together with independence in the occurrence of the BCD type from a missing $Y(1)$ value, MAR-2 represents the main effects of $Y(2)$ and $Y(3)$ on the distinction of the ABCD, ABC, and BCD types, MAR-3 represents the dependence of the distinction of the ABCD and BCD types on $Y(4)$ together with independence in the occurrence of the ABC type from a missing $Y(4)$

TABLE 5
Results from 15 Models

	L-Squared	df	BIC
1. Two latent classes			
MCAR	431.92	851	-4,632.17
MAR-1	430.46	847	-4,600.86
MAR-2	420.96	835	-4,539.07
MAR-3	428.47	847	-4,602.04
NIR	429.17	849	-4,613.73
2. Three latent classes			
MCAR	349.77	846	-4,675.61
MAR-1	348.32	842	-4,653.30
MAR-2	338.83	830	-4,591.51
MAR-3	346.32	842	-4,655.30
NIR	347.50	842	-4,654.12
3. Four latent classes			
MCAR	343.41	841	-4,652.26
MAR-1	342.83	837	-4,629.10
MAR-2	332.44	825	-4,568.20
MAR-3	339.78	837	-4,632.14
NIR	332.85	835	-4,627.20

TABLE 6
Results from the Model with Three Latent Classes and MCAR

	1st–2nd Year	3rd–4th Year	5th–6th Year	7th–8th Year
1. Latent class 1: Size = 19.1%				
Very satisfied	0.704	0.741	0.913	0.620
Pretty satisfied	0.280	0.255	0.086	0.372
Neutral	0.015	0.004	0.001	0.018
Somewhat dissatisfied	0.001	0.000	0.000	0.000
Not satisfied	0.000	0.000	0.000	0.000
2. Latent class 2: Size = 57.0%				
Very satisfied	0.351	0.167	0.108	0.063
Pretty satisfied	0.527	0.707	0.584	0.515
Neutral	0.110	0.123	0.278	0.331
Somewhat dissatisfied	0.011	0.003	0.029	0.082
Not satisfied	0.001	0.000	0.001	0.009
3. Latent class 3: Size = 23.9%				
Very satisfied	0.072	0.001	0.005	0.012
Pretty satisfied	0.415	0.102	0.156	0.252
Neutral	0.332	0.518	0.471	0.409
Somewhat dissatisfied	0.127	0.313	0.315	0.254
Not satisfied	0.054	0.066	0.053	0.073

value, and NIR represents the dependence of the three missing-data types on the latent-class variable X .⁷ (See Appendix A for applying these models by the use of LEM.) The results from Table 5 indicate that the latent class model that hypothesizes three latent classes and MCAR attains the best fit with the data according to both the LL test and BIC.

Table 6 presents the results from this best-fitting model. The results show that about 19 percent of women in latent class 1 have a trajectory of high satisfaction levels, with “very satisfied” with the marriage as a modal response. The level of their satisfaction tends to increase up to the 5th–6th year of marriage but somewhat declines in the 7th–8th year. About 57 percent of women have a middle-level trajectory of marital satisfaction but also experience a steady decline in satisfaction as marriage duration increases, with the probability of “very satisfied” or “pretty satisfied” being high in the 1st–2nd year and the probability of

⁷ It is technically difficult to combine all of MAR-1, MAR-2, and MAR-3. We can, however, combine MAR-1 and MAR-3. The results, which are not significantly better than MCAR, are omitted, however.

“pretty satisfied” or “neither satisfied nor dissatisfied” being high in the 7th–8th year. About 24 percent have a relatively low level trajectory, and their dissatisfaction level decreases up to the 5th–6th year of marriage, where there is about a one-third probability of being dissatisfied, but the satisfaction level somewhat improves in the 7th–8th year.

The second step of the analysis first applies the latent-trajectory model by including three time-constant covariates for the composition of latent classes. The covariates are all measured at the time that the subject’s marital satisfaction was first measured. The covariates are (1) wife’s employment status (full-time homemakers versus employed wives), (2) wife’s confidence in husband’s family-financing capacity, and (3) wife’s confidence in husband’s emotional support. Husband’s income was also considered as an additional covariate but was omitted because its unique effects become nonsignificant when wife’s confidence in husband’s family-financing capacity is controlled. Table 7 gives the descriptive statistics of the three covariates.

The introduction of these covariates affects the trajectory pattern of each latent class very little, although the probability of being in each latent class now varies with covariates. Table 8 presents results from Models 1 and 2. The LEM program for model 2 is given in Table A3 in Appendix A. Model 1 includes only the main effects of the three

TABLE 7
Distributions of Covariate States: *N* = 380

	Proportion (%)
1. Employment Status	
Fulltime homemaker	57.3
Employed	42.7
2. Confidence in husband’s family-financing capacity	
High	29.2
Somewhat high	31.6
Average	34.5
Somewhat low	2.9
Low	1.8
5. Confidence in husband’s emotional support	
High	55.5
Somewhat high	23.9
Average	16.6
Somewhat low	2.9
Low	1.1

TABLE 8
Results from Latent-Trajectory Models¹

	Model 1		Model 2	
1. Covariate effects on $\log(\pi_{L_i}^X/\pi_{L_2}^X)$	L1	L3	L1	L3
	versus L2	versus L2	versus L2	versus L2
(1) Employment status (fulltime homemaker versus employed)	-0.236	-0.256	4.479*	2.233
(2) Confidence in husband's family-financing capacity	0.593*	-0.778**	1.209**	-0.394
(3) Interaction effects: (1) x (2)	—	—	-1.110*	-0.697
(4) Confidence in husband's emotional support	1.989*	-1.451***	2.135*	-1.462***
2. The Effects of L1 versus L2				
on $\log(\pi_j^{Y_1}/\pi_{j+1}^{Y_1})$	1.792**		1.822**	
on $\log(\pi_j^{Y_2}/\pi_{j+1}^{Y_2})$	2.929***		2.823***	
on $\log(\pi_j^{Y_3}/\pi_{j+1}^{Y_3})$	3.021***		3.104***	
on $\log(\pi_j^{Y_4}/\pi_{j+1}^{Y_4})$	2.367***		2.489***	
3. The effects of L3 versus L2				
on $\log(\pi_j^{Y_1}/\pi_{j+1}^{Y_1})$	-1.810***		-1.841***	
on $\log(\pi_j^{Y_2}/\pi_{j+1}^{Y_2})$	-3.777***		-3.701***	
on $\log(\pi_j^{Y_3}/\pi_{j+1}^{Y_3})$	-1.513***		-1.526***	
on $\log(\pi_j^{Y_4}/\pi_{j+1}^{Y_4})$	-0.801***		-0.804***	
Log-likelihood	-2,472.36		-2,468.85	
Number of parameters ²	34		36	

¹L1: Latent class with high-level trajectory.
L2: Latent class with middle-level trajectory.
L3: Latent class with low-level trajectory.

²Sixteen parameter estimates for the intercepts of indicators, two parameter estimates for the intercept of the latent class regression, and two parameter estimates for the missing-indicator variable are omitted.

***p < .001; **p < .01; *p < .05.

covariates on the composition of latent classes and model 2 adds to model 1 the interaction effects of wife's employment status and confidence in husband's family-financing capacity, which show significance. The results on the effects of the latent class variable on $Y(t)$, for $t = 1 - 4$, show that latent class 1 has consistently higher level of marital satisfaction than latent class 2, and latent class 3 has consistently lower level of marital satisfaction than latent class 2. We can therefore see

that latent classes 1, 2, and 3 are “high-level trajectory,” “middle-level trajectory,” and “low-level trajectory” classes, respectively.

The results from model 1 show that a wife’s employment status does not affect the latent class composition, and greater confidence in husband’s family-financing capacity and greater confidence in husband’s emotional support both increase the odds of being in the high-level trajectory class versus the middle-level trajectory class, and decrease the odds of being in the low-level trajectory class versus the middle-level trajectory class. The results from model 2, however, show that, among employed wives, greater confidence in husband’s family-financing capacity significantly increases the odds of being in the high-level trajectory class versus the middle-level trajectory class (1.209) but does not affect the odds of being in the low-level trajectory class versus the middle-level trajectory class ($0.099 = 1.209 - 1.110$), while among wives who are full-time homemakers, greater confidence in husband’s family-financing capacity significantly decreases the odds of being in the low-level trajectory class versus the middle-level trajectory class ($-1.091 = -0.394 - 0.697$, which is significant at the 5 percent level) but does not affect the odds of being in the high-level trajectory class versus the middle-level trajectory class (-0.394). Table 9 reveals these characteristics of interaction effects by showing how the probabilities of being in latent class 1 (high-level trajectory) and latent class 3 (low-level trajectory) change as a function of the covariates, and the results also show great variability in the composition of latent classes with the covariates.

The results from the adjacent logit latent growth curve models applied to the same data are now presented. Although the details of the statistics involved in model selection are omitted, the best model was selected by a procedure based on the likelihood ratio test. Four categories of effects are tested for significance in the following order: (1) covariate effects on the “intercept,” (2) covariate effects on the “slope,” (3) interaction effects of covariates and the latent-class variable on the “intercept,” and (4) interaction effects of covariates and the latent-class variable on the “slope.” Except for step 1, nonsignificant effects are omitted from the model before proceeding to the next step, mainly because the small size means that the results are more stable without nonsignificant effects. In addition, at each step, models with one class (no unobserved population heterogeneity), two classes, and three classes are tested and compared. The interaction effects among covariates are also tested in the first step. The resulting best-fitting model was still a three-class model,

TABLE 9
Predicted Probabilities of Being in Latent Classes 1 and 3: Results from Model 2

		Proportion of Latent Class 1 (High-Satisfaction Class)			
		Full-time Homemaker Family-Financing Capacity		Employed Wives Family-Financing Capacity	
		High	Somewhat high	Average or below	
Emotional support	High	.32	.29	.25	High
	Somewhat high	.05	.04	.03	Somewhat high
	Average or below	.01	.00	.00	Average or below
					High
					Somewhat high
					Average or below
		Proportion of Latent Class 3 (Low-Satisfaction Class)			
		Full-time Homemakers Family-Financing Capacity		Employed Wives Family-Financing Capacity	
		High	Somewhat high	Average or below	
Emotional support	High	.01	.04	.11	High
	Somewhat high	.07	.19	.41	Somewhat high
	Average or below	.26	.51	.76	Average or below
					High
					Somewhat high
					Average or below

as in the results of the analysis in the previous section. Table 10 presents the best mixed model (where the “intercept” and the “slope” do not vary with latent classes), and the best mixture model identified by this procedure. Table A4 in Appendix A gives the LEM program for model 2 in Table 10.

TABLE 10
Results from Latent Growth Curve Models¹

	Model 1	Model 2
1. Wife’s employment status (full-time homemakers versus employed)		
On intercept	−0.251	−0.106
On slope	0.220*	0.175†
2. Confidence in husband’s family-financing capacity		
On intercept	0.590***	0.538***
On slope	−0.118*	−0.102†
3. Confidence in husband’s emotional support		
On intercept	1.491***	1.026***
On slope	−0.331***	−0.297***
4. Interaction effects: husband’s emotional support x latent class on the slope		
L1 versus L2	—	1.544***
L3 versus L2	—	−0.081
5. The effects of L1 versus L2		
On $\log(\pi_j^{Y_1}/\pi_{j+1}^{Y_1})$	0.484	−1.049*
On $\log(\pi_j^{Y_2}/\pi_{j+1}^{Y_2})$	1.977***	−0.771†
On $\log(\pi_j^{Y_3}/\pi_{j+1}^{Y_3})$	2.990***	−0.293
On $\log(\pi_j^{Y_4}/\pi_{j+1}^{Y_4})$	1.973***	0.389
6. The effects of L3 versus L2		
On $\log(\pi_j^{Y_1}/\pi_{j+1}^{Y_1})$	−1.326**	−0.775
On $\log(\pi_j^{Y_2}/\pi_{j+1}^{Y_2})$	−2.746***	−2.389***
On $\log(\pi_j^{Y_3}/\pi_{j+1}^{Y_3})$	−2.363***	−2.136***
On $\log(\pi_j^{Y_4}/\pi_{j+1}^{Y_4})$	−1.448***	−1.186*
Latent class compositions		
L1	0.261	0.264
L2	0.634	0.611
L3	0.106	0.125
Log-likelihood	−2,444.71	−2,440.93
Number of parameters ²	34	36

¹L1: Latent class with high satisfaction-level trajectory.
L2: Latent class with middle satisfaction-level trajectory.
L3: Latent class with low satisfaction-level trajectory.
²Sixteen parameter estimates for the intercepts of indicators, two parameter estimates for the intercept of the latent class regression, and two parameter estimates for the missing-indicator variable are omitted.
***p < .001; **p < .01; *p < .05; †p < .10.

Results from model 1 in Table 10 do not change the fundamental characteristics of the three latent classes. As shown in the effects of the latent class variable on $Y(t)$ in Table 8, class 1 is the high-level trajectory class, class 2 is the middle-level trajectory class, and class 3 is the low-level trajectory class. Covariate effects in Table 8 show that while full-time homemakers tend to have a lower level of marital satisfaction (although the difference is not significant) in the first two years of marriage, the slope of its growth is significantly higher—which in fact implies that the tendency for marital satisfaction to decrease with marital duration (characterized by nonlinear time effects) is slower—for wives who are full-time homemakers than for employed wives, thereby reversing the order of the relative satisfaction between the two groups. The effects of both higher confidence in husband's family financing capacity and higher confidence in husband's emotional support measured early in marriage on increasing marital satisfaction attenuates as marriage duration increases, as indicated by a positive effect on the intercept combined with a negative effect on the slope of growth.

Model 2 in Table 10 indicates that the effects of confidence in husband's emotional support on the intercept differ among the three latent classes, and the effects for latent class 1 are much greater than those for the other two latent classes. Note that when we have the interaction effects of the latent class variable and a covariate, there are more than three distinct latent trajectories because the trajectory pattern also changes with the covariate. Although the variable for confidence in husband's emotional support has five categories, the fourth and fifth categories are small (see again Table 7), and therefore we may consider that we have nine representative trajectories as a combination of the latent classes and the covariate states.

Figure 1 depicts how the logit of having one higher level of marital satisfaction varies with the combination of the three covariate states and the three latent classes at each marital duration time point—by holding threshold-specific main nonlinear time effects constant and by setting the state of latent class 2 for people who have an “average” level of confidence in husband's emotional support at the 0 “baseline” state for comparison.⁸ Since the change in time in Figure 1 does not take into

⁸ Confidence in husband's emotional support here is based on a coding of 2, 1, 0, -1, -2 respectively for “high,” “somewhat high,” “average,” “somewhat low,” and “low” levels of confidence.

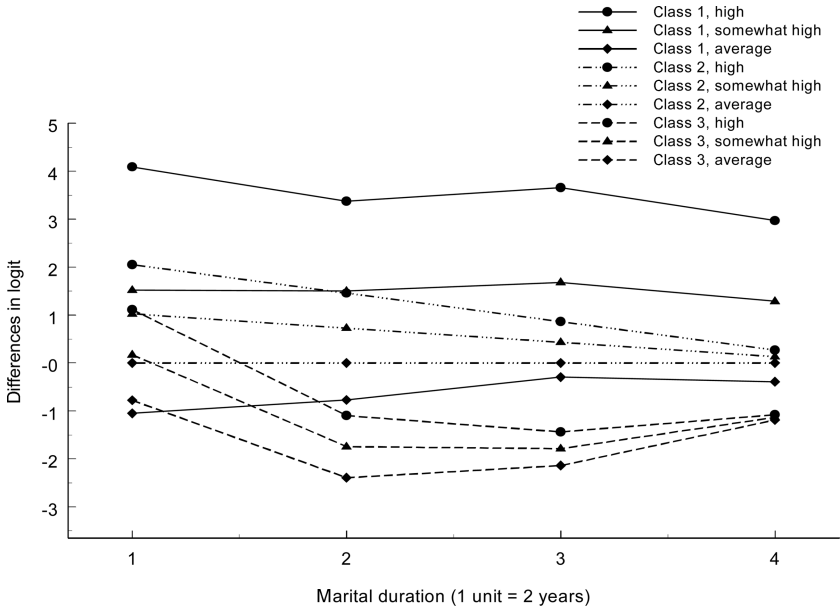


FIGURE 1. Differences in logit among nine groups.

account the threshold-specific main time effects that apply equally to all these nine groups, the lines in the figure should be compared only vertically at each time point.

Figure 1 shows that latent class 1 is not only a group whose average level of marital satisfaction is higher than that of the other groups but also a group in which the difference in confidence in husband's emotional support has a much more lasting influence on the level of marital satisfaction than it does in the other two classes. This is because the initial differences in the extent of marital satisfaction among those who differ in this confidence level are much greater in latent class 1 than the initial differences in the other two latent classes. As discussed in Section 6, an increase in the effects of the latent class variable on marital satisfaction with confidence in husband's emotional support also indicates that subjects with higher confidence in husband's emotional support tend to have higher stability in their marital satisfaction.

Finally, since the number of parameters employed in models 1 and 2 in Table 10 are the same as those of the latent-trajectory models 1 and 2 in Table 8, respectively, the fact that the log-likelihood values

of the models in Table 10 are greater than those of the models in Table 8 shows that the latent growth curve models have a greater explanatory power than the latent trajectory models for the data analyzed here. However, substantive findings that can be made from these two models are somewhat different, and therefore, this statistical criterion alone may not determine which model should be chosen. Researchers may make a choice between the latent trajectory model and the latent growth curve model depending on substantive issues to be analyzed.

9. CONCLUSION AND FURTHER REMARKS

This paper has presented two distinct and useful ways to specify the mixture Markov model for regression analysis of panel data with a categorical dependent variable. The specification is employed because (1) parameter estimates from a general mixture Markov model are usually very unstable, and (2) the joint distribution of $Y(t)$ and $Y(t - 1)$ has two structural components—namely, the trajectory component which characterizes change in the marginal distribution of $Y(t)$, and the state dependence component that characterizes the association between $Y(t)$ and $Y(t - 1)$ independently of the marginal distributions. Two novel mixture Markov regression models, the mover-random switching regression model and the mover-stayer regression model, which assume unobserved population heterogeneity in state dependence, are introduced for simultaneous analysis of response probability and response stability. While the response stability can be analyzed in different ways, the use of a latent class variable as an indicator of unobserved heterogeneity in response stability permits an analysis of response probability and response stability in a simultaneous regression framework, which can be quite informative.

Two other regression models that assume unobserved heterogeneity in the trajectory of the dependent variable are also introduced. One of them, the adjacent logit latent trajectory model, is a special case of the multinomial logit latent class regression model, and its advantage over the alternative adjacent logit latent growth curve model is its simplicity of model selection and a description of distinct trajectories and their associations with covariates. Its major disadvantage is a lack of insight into covariate effects on “growth” patterns beyond what can be

captured by a mixture of latent trajectory patterns, because the model assumes covariate effects only on the composition of latent classes. The most significant advantage of the adjacent logit latent growth curve model is its capacity to reflect the dependence of trajectories, and/or growth, on covariates, latent classes, and their combinations. There are, however, relative costs associated with the use of this model. The model is more demanding of data and information, model selection becomes more complicated, partly because we must identify the optimal number of latent classes for each given set of covariate effects, and the interpretation of the model's results may also become more complicated than in the latent trajectory model. However, if information in the sample is sufficiently rich, this alternative may provide a deeper insight into the data than the latent trajectory model.

While latent trajectory and latent growth curve models are introduced for the analysis of a categorical dependent variable with ordered categories, the two models for the analysis of response stability can be employed for a categorical dependent variable with or without ordered categories. For a dependent variable with ordered categories, the use of the mover-random model may benefit from an elaboration in modeling the dependence of $Y(t)$ on $Y(t - 1)$ among movers by using association models (Goodman 1979) that take into account an order among categories. This elaboration, however, is not required, and the dependence of $Y(t)$ on $Y(t - 1)$ among movers may be left unstructured. In the mover-stayer model, however, we must employ association models to characterize the *off-diagonal* association between $Y(t)$ and $Y(t - 1)$ among movers. In such applications, the characteristics of the mover-random and mover-stayer models become quite different because off-diagonal association between $Y(t)$ and $Y(t - 1)$ is represented by a latent class with greater response stability in the mover-random model, while it is represented by a latent class with smaller response stability in the mover-stayer model. The stayer-mover-random model, which combines the mover-random and mover-stayer models, will also be useful for a dependent variable with ordered categories. On the other hand, the stayer-random model is usually inadequate for a dependent variable with ordered categories, because $Y(t)$ usually depends on $Y(t - 1)$ beyond that reflected by the presence of stayers.

This paper has also presented illustrative applications of these four models with panel data providing additional guidance for the use of LEM for applications.

APPENDIX A: SAMPLE LEM PROGRAMS

Table A1 presents the LEM program used for mover-random model III-A in Table 2. The `man` statement specifies the number of manifest (or observed) variables and the `lat` statement specifies the number of latent variables. The `label (lab)` statement specifies variable labels, and label X is employed for the latent-class variable, labels A, B, C are employed for $Y(t)$, for $t = 1, 2$, and 3 respectively, labels R through V are employed for covariates as described in a comment in the program, and the `dim` statement indicates the number of their categories. The `model (mod)` statements specify three simultaneous equations for X (the latent variable), $B (= Y(2))$, and $C (= Y(3))$ and how they depend in logit on covariates. The `design (des)` statements are used here to impose the commonality of covariate effects between the equation for predicting B and that for predicting C, by statements following `all` (which specify that a pair of interaction terms share the same coefficients) in the `model` statement. The deviation contrast is employed for A, B, and C by specifying -1 , and an indicator contrast (with a set of dummy variables) is employed for other variables by specifying the omitted category in the `dum` statement. See Vermunt (1997b) for more information about the model and design statements. The program assumes cross-classified frequency data, labeled `pub12.fre` here, as input data. Since latent variable X is coded by a dummy variable of 1 versus 0, the absence of AB and BC interactions, together with the presence of XAB and XBC interactions, characterizes the mover-random model with the state-dependence effect of Y as the separation indicator.

Table A2 presents the LEM program used for the mover-stayer model III-B1 in Table 2. The labels of observed variables are the same as those in Table A1. Latent class variable X makes a dichotomous distinction between stayers, for whom $Y(1) = Y(2) = Y(3)$ holds, and movers, for whom $Y(t)$ depends on $Y(t - 1)$. The characteristics of stayers for $X = 1$ are specified by imposing “structural zeros” in the starting weight statements `sta wei(XAB)` and `sta wei(XBC)` on cases where $A \neq B$ or $B \neq C$ hold. The statement `sta wei(XAB) [1 0 0 1 1 1 1 1]`, for example, imposes a zero probability of occurrence for two cases—the case where $X = 1$, $A = 1$, and $B = 2$, and the case where $X = 1$, $A = 2$, and $B = 1$ —among the eight combinations of the values of X, A, and B. See the LEM manual (Vermunt 1997b) for the use of the starting weight statements. The term `cov(AB, BC, 1)` in the model

TABLE A1
The LEM Program for the Mixed Markov Mover-Random Regression Model for
the Analysis of Response Stability

lat 1	
man 8	
dim 2 2 2 2 5 2 4 4 3	
lab X A B C R S T U V *A:Y1 B:Y2 c:Y3 R:age;S:race;	
*T:educ;U:rel;V:party	
mod X ARSTUV {X,XR,XS,XT,XU,XV,XA}	
B XARSTUV {B}	
C XBRSTUV {C}	
all {cov(XB,XC,1),cov(XAB,XBC,1),cov(BR,CR,4),	
cov(BS,CS,1),cov(BT,CT,3),	
cov(BU,CU,3),cov(BV,CV,2)}	
*Note: There are no main effects of {AB, BC}	
interactions	
*In other words, lagged effects exist only	
for X = 1	
dum 2 -1 -1 -1 1 1 1 1 1	
des [1 -1 0 0	*cov(XB,XC,1)
1 -1 0 0	
1 -1 -1 1 0 0 0 0	*cov(XAB,XBC,1)
1 -1 -1 1 0 0 0 0	
0 1 0 0 0 0 -1 0 0 0	*cov(BR,CR,4)
0 0 1 0 0 0 0 -1 0 0	
0 0 0 1 0 0 0 0 -1 0	
0 0 0 0 1 0 0 0 0 -1	
0 1 0 0 0 0 -1 0 0 0	
0 0 1 0 0 0 0 -1 0 0	
0 0 0 1 0 0 0 0 -1 0	
0 0 0 0 1 0 0 0 0 -1	
0 1 0 -1	*cov(BS,CS,1)
0 1 0 -1	
(The rest of the design statements are omitted.)	
]	
nfr	
dat publ2.fre	

statements specifies a common parameter for the dependence of Y(2) on Y(1) and of Y(3) on Y(2) among movers, as described in the design statements.

Table A3 presents the LEM program used in model 2 of Table 4 for the latent trajectory model. The res statement indicates the presence of missing data and the number of missing indicators. Label R is

TABLE A2
The LEM Program for the Mixed Markov Mover-Stayer Regression Model

```

lat 1
man 8
dim 2 2 2 2 5 2 4 4 3
lab X A B C R S T U V *A:Y1 B:Y2 c:Y3 R:age;S:race;
    *T:educ;U:rel;V:party
mod X|ARSTUV {X,XA,XR,XS,XT,XU,XV,XAV}
    B|XARSTUV {B,wei(XAB)}
    C|XBRSTUV {C,wei(XBC)}
    all {cov(AB,BC,1),cov(BR,CR,4),cov(BS,CS,1),
        cov(BT,CT,3),cov(BU,CU,3),cov(BV,CV,2)}
dum 2 2 2 2 1 1 1 1 1
des [1 0 0 0                *cov(AB,BC,1)
    1 0 0 0
    0 1 0 0 0 0 0 0 0      *cov(BR,CR,4)
    0 0 1 0 0 0 0 0 0
    0 0 0 1 0 0 0 0 0
    0 0 0 0 1 0 0 0 0
    0 1 0 0 0 0 0 0 0
    0 0 1 0 0 0 0 0 0
    0 0 0 1 0 0 0 0 0
    0 0 0 0 1 0 0 0 0
    0 1 0 0                *cov(BS,CS,1)
    0 1 0 0
    ]
    sta wei(XAB) [1 0 0 1 1 1 1 1]
    sta wei(XBC) [1 0 0 1 1 1 1 1]
    nfr
dat publ2.fre

```

employed for the single missing indicator, label X is employed for the latent class variable, labels A, B, C, and D are employed here for $Y(t)$, for $t = 1, 2, 3$, and 4, respectively, and labels E, P, and Q are employed for covariates as described in a comment in the program.

The sub statement identifies the three types of observable cross-classified frequencies with missing data. Although we have ABCD, ABC, and BCD as the three types, they must be given as cross-classifications with covariates as specified in the statement.

In the model statement, the specification $\{A, \text{cov}(AX, 2)\}$ for the dependence of A (= $Y(1)$) on X, $A|X$, and similar specifications for $B|X$, $C|X$, and $D|X$, indicate that the marginal distribution for

TABLE A3
The LEM Program for the Latent Trajectory Regression Model

```
man 7
lat 1
res 1
dim 3 3 5 5 5 5 2 5 5
lab R X A B C D E P Q *A:Y1;B:Y2;C:Y3;D:Y4:
                                *E:employment status
                                *P:family-finance capacity;
                                *Q:emotional support
sub BCDEPQ ABCDEPQ ABCEPQ
mod X|EPQ {X,cov(XE,2), cov(XP,2),cov(XQ,2), cov(XEP,2)}
  A|X {A,cov(AX,2)}
  B|X {B,cov(BX,2)}
  C|X {C,cov(CX,2)}
  D|X {D,cov(DX,2)}
  R
rec 380
mis 0
des [1 0 0 0 0 0 *cov(XE,4)
    0 0 1 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0 *cov(XP,2)
    0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0 *cov(XQ,2)
    0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0
    0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 *cov(XEP,2)
    0 0 0 0 0 0 0 0 0 0 2 1 0 -1 -2
    0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0 *cov(AX,2)
    0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0 *cov(BX,2)
    0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0 *cov(CX,2)
    0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0
    2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0 *cov(DX,2)
    0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0]
dat ms16.dat
```

each $Y(t)$ is fixed and the effects of the latent-class variable X on $Y(t)$, characterized by two parameters described below, vary with time t .

The entry R in the model statement indicates MCAR. If we instead hypothesize that the missing data are not missing at random but depend on latent trajectory patterns, this needs to be replaced with $R|X$

{R, RX}. Similarly, an alternative specification as $R|BC$ {R, RB, RC} yields the MAR-2 model where the missing data depend on the value of B (= Y(2)) and C (= Y(3)). A specification for MAR-1, where it is assumed that the missing data of A (= Y(1)) occur randomly in the BCD type but the distribution of A varies with the distinction of the ABCD and ABC types, requires equality constraints on the conditional probabilities of R for a given A, and can be specified as $R|A$ eq2 in the model statement combined with the entry 0 0 0 0 0 0 0 0 0 1 1 1 1 1 in the design statement. See the LEM manual for the use of eq2 for equality constraints.

The rec statement indicates the input of individual data and the number of individual records there. The mis 0 statement indicates that the missing data have a nominal code of 0 in the input data. Note that the number of categories in the dim statement does not count missing cases as a separate category, and all nonmissing cases must have codes other than 0.

The design statement specifies the category-by-linear interactions for elements AX, XB, XC, and XD by imposing a standardized linear scale (2, 1, 0, -1, -2) for A through D, and this yields the proportional odds effects of X on Y(t). Indicator contrasts with the last category as the omitted category are employed for X and for covariates E, P, and Q regarding their effects on X. The use of standardization for the linear contrast of Y(t) so that the scale has a mean of 0 is strongly recommended for both the adjacent logit latent trajectory model and the adjacent logit latent growth curve model in order for the iterative search for the maximum likelihood estimates to be successful.

Table A4 presents the LEM program for model 2 in Table 7 for the latent growth curve model. The model statement specifies that Y(t) depends on the latent class variable X and covariates E, P, and Q. The specification for all in the model statements indicates common regression coefficients. For example, cov(AE,BE,CE,DE,2) indicates that two parameters commonly characterize the effects of employment status (E) on Y(t) (A, B, C, and D), and the corresponding design statements specify that one parameter is the effect of E (1 versus 0) on the linear contrast of Y(t) scaled as (2,1,0,-1,-2), and the other is the interaction effect tE on Y(t), where the time variable t is coded as 0, 1, 2, and 3 for A, B, C, and D, respectively. This pair of parameters represents the effects of this covariate on the “intercept” and the “slope” of growth on the logit. The dependence of the effects of covariate Q on

TABLE A4
The LEM Program for the Switching Regression Model for the Growth-Curve Analysis

man 7	
lat 1	
res 1	
dim 3 3 5 5 5 5 2 5 5	
lab R X A B C D E P Q	
sub BCDEPQ ABCDEPQ ABCEPQ	
mod X	
A XEPQ {A,cov(AX,2)}	
B XEPQ {B,cov(BX,2)}	
C XEPQ {C,cov(CX,2)}	
D XEPQ {D,cov(DX,2)}	
R	
all {cov(AE,BE,CE,DE,2),cov(AP,BP,CP,DP,2),	
cov(AQ,BQ,CQ,DQ,2)cov(XAQ, XBQ, XCQ, XDQ,2)}	
rec 380	
mis 0	
des [2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0	*cov(AX,2)
0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0	
2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0	*cov(BX,2)
0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0	
2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0	*cov(CX,2)
0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0	
2 1 0 -1 -2 0 0 0 0 0 0 0 0 0 0	*cov(DX,2)
0 0 0 0 0 2 1 0 -1 -2 0 0 0 0 0	
2 0 1 0 0 0 -1 0 -2 0	*cov(AE,2)
0 0 0 0 0 0 0 0 0 0	
2 0 1 0 0 0 -1 0 -2 0	*cov(BE,2)
2 0 1 0 0 0 -1 0 -2 0	
2 0 1 0 0 0 -1 0 -2 0	*cov(CE,2)
4 0 2 0 0 0 -2 0 -4 0	
2 0 1 0 0 0 -1 0 -2 0	*cov(DE,2)
6 0 3 0 0 0 -3 0 -6 0	
4 2 0 -2 -4 2 1 0 -1 -2 0 0 0 0 0 -2 -1	
0 1 2 -4 -2 0 2 4	
*cov(AP,1)	
(The rest of the design statements are omitted.)	
]	
nfr	
dat ms16.dat	

latent classes is specified by $\text{cov}(\text{XAQ}, \text{XBQ}, \text{XCQ}, \text{XDQ}, 2)$ and the corresponding design statements that are omitted in Table A4.

APPENDIX B: NOTES ON THE USE OF TIME-DEPENDENT COVARIATES

All models with a latent class variable introduced in this article can employ time-dependent covariates and can be applied to data by using the LEM program. An important caveat should be noted here: for a model that uses $Y(t)$ and time-dependent covariates $V(t)$, we should employ neither the person-period data nor the cross-classified frequency data that include time variable T as one dimension of cross-classification. The use of such data sets yields time dependence for the latent class variable, which is not assumed in the model. Instead, we should analyze either individual data that attach $V(t)$ measured at different times as a set of independent variables or cross-classified frequency data that treat each time-dependent covariate by time as a set of covariates. In other words, in defining the latent-class variable, we should treat $Y(t - 1)$ and $V(t)$ for each t in predicting $Y(t)$ as if they were separate time-constant variables. Suppose that A, B, and C are LEM labels for $Y(t)$ at times 1, 2, and 3, and R and S are LEM labels for $V(t)$ at times 2 and 3. Then the inclusion of $\text{cov}(\text{AB}, \text{BC}, m)$ and $\text{cov}(\text{BR}, \text{CS}, n)$ under the all specification of the model statement in LEM indicates the use of m parameters that commonly characterize the dependence of $Y(t)$ on $Y(t - 1)$ for $t = 2$ and 3, and the use of n parameters that commonly characterize the dependence of $Y(t)$ on $V(t)$ for $t = 2$ and 3, respectively.

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